

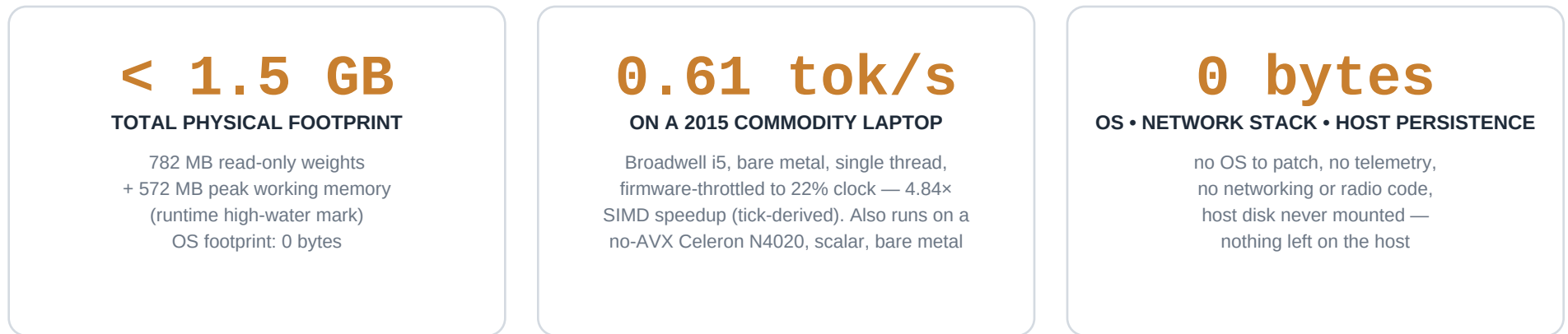
A.L.I.C.E. — Sovereign Bare-Metal AI for DDIL Operations

DARPA-SN-26-97 Low Resource Computing | 1(b) Low Memory + 2(d) Low Complexity of UX

ARCHITECTURE — THE OPERATING SYSTEM IS DELETED, NOT OPTIMIZED



MEASURED ON REAL LEGACY HARDWARE — EVERY NUMBER RUNTIME-COMPUTED AND LOGGED



ROADMAP: model portfolio to sub-1 GB (Falcon-E port in progress) → distilled DDIL mission model (published BitDistill recipe) → embedded x86 SoC targets

Compatibility floor (assessed): 64-bit UEFI (~2012+), Secure Boot off (one firmware toggle; signed boot on roadmap), 2 GB RAM — bracketed under QEMU: 2 GB generates, 1.4 GB explicit OOM. Provenance: bare-metal figures runtime-computed by the engine's /gauntlet instrumentation on 3 machines (Broadwell AVX2, Celeron N4020 scalar, QEMU matrix); ratios from raw TSC ticks; all raw logs in the versioned repository (hashes on request). Quality: WikiText-2 perplexity 15.8 full test set, host-side harness, pruned vocabulary — caveats logged with the number.

Model: Microsoft BitNet b1.58-2B-4T (open, MIT). Systems contribution: the no_std bare-metal inference stack. Engine performs no targeting, no fire control, no transmission.